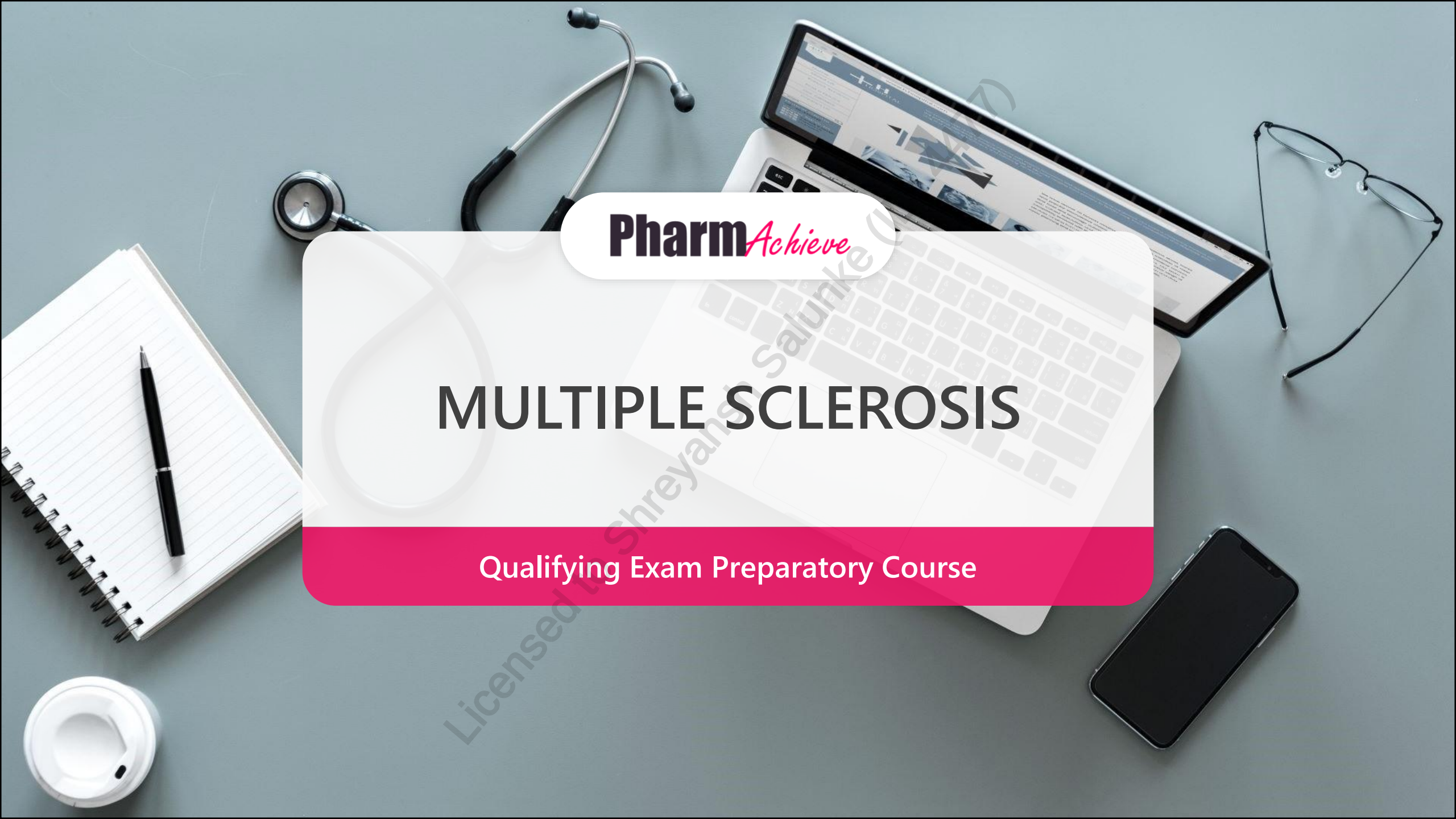


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a white coffee cup. A large, semi-transparent white pill shape is centered over the laptop.

Pharm*Achieve*

MULTIPLE SCLEROSIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ANA Anti-Nuclear Antibody

BCRP Breast Cancer Resistance Protein

CBC Complete Blood Count

CI Contraindicated

CIS Clinically Isolated Syndrome

CRP C-Reactive Protein

DMT Disease-Modifying Therapies

EDSS Expanded Disability Status Scale

GI Gastrointestinal

HA Headache

IV Intravenous

JCV John-Cunningham Virus

LFT Liver Function Test

MAOI Monoamine Oxidase Inhibitor

MS Multiple Sclerosis

PML Progressive Multifocal Leukoencephalopathy

PPMS Primary Progressive Multiple Sclerosis

RRMS Relapsing-Remitting Multiple Sclerosis

RF Rheumatoid Factor

S1P Sphingosine 1-Phosphate

SC Subcutaneous

SE Side Effects

SPMS Secondary Progressive Multiple Sclerosis

URTI Upper Respiratory Tract Infections

DEFINITIONS

Gait	The way a person walks
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Multiple Sclerosis (MS)	A chronic neurologic disease in which the body's immune system damages its own nerve cells and connections between the brain and spinal cord
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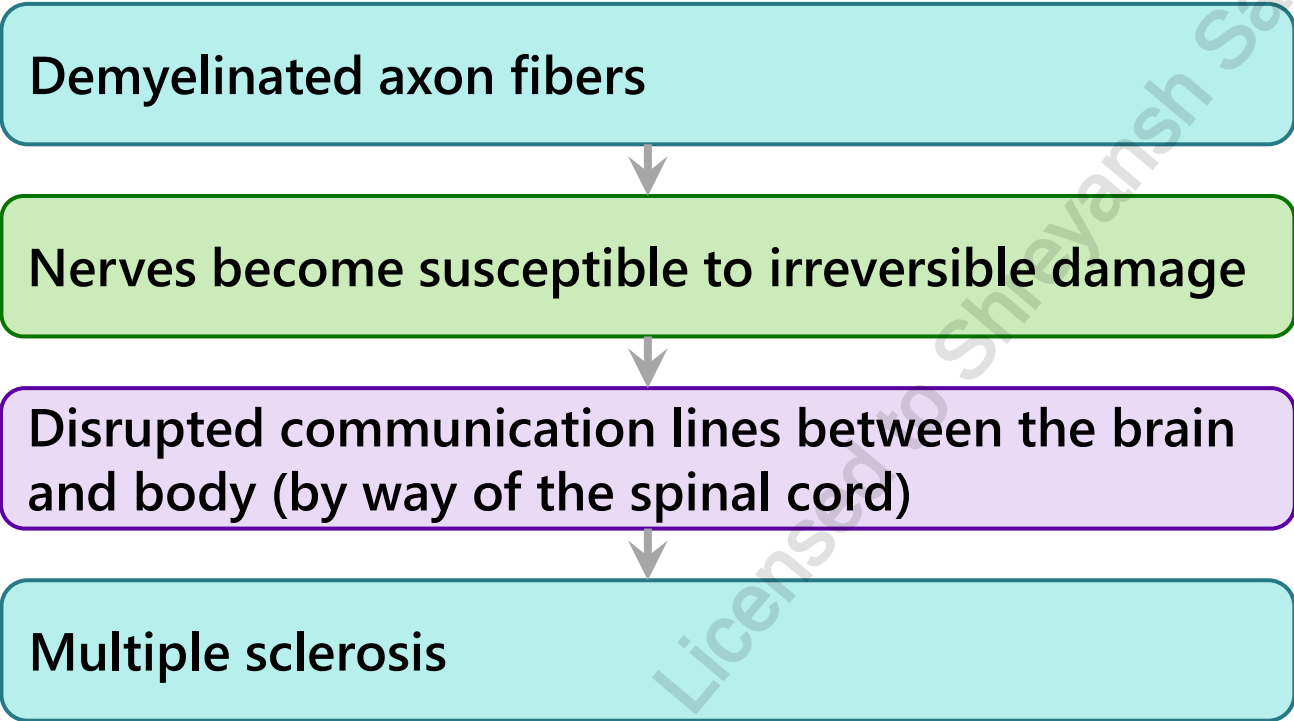
Myelin	A fatty protective layer that covers the axon fibers, that protects the nerves in the brain and spinal cord
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PATHOPHYSIOLOGY

The triggers of MS are **not fully understood**, but a postulated theory is

1. Body has an immune response to a foreign molecule similar in structure to myelin
2. Local inflammation occurs
3. Demyelination of axons occurs

Demyelinated axon fibers



```
graph TD; A[Demyelinated axon fibers] --> B[Nerves become susceptible to irreversible damage]; B --> C[Disrupted communication lines between the brain and body (by way of the spinal cord)]; C --> D[Multiple sclerosis];
```

Nerves become susceptible to irreversible damage

Disrupted communication lines between the brain and body (by way of the spinal cord)

Multiple sclerosis

RISK FACTORS

Female sex (2-3x more likely)

Family history in 1st degree relatives

Variations in HLA genes (e.g HLA-DRB1*1501)

Viral infections (measles, Epstein-Barr, human herpes)

Changes in gut microbiome

Deficiency of vitamin D (especially in pregnancy or in neonates)

Smoking

Obesity

Certain autoimmune diseases

CLINICAL PRESENTATION

Vision issues (including possible vision loss)

Dizziness

Bladder and/or bowel dysfunction

Dysesthesias

Numbness and weakness of limbs

Ataxia

Neuropathic pain

Tremors

Lhermitte's Phenomenon

Uhthoff Phenomenon

Presentation is highly variable
and patient-specific

MEDICAL/DRUG RELATED CAUSES

Vitamin B 12 Deficiency may mimic symptoms

TNF-alpha inhibitors are associated with an increased risk of MS

Need to rule out all other causes prior to diagnosis

THE 4 SUBTYPES OF MULTIPLE SCLEROSIS

Clinically Isolated Syndrome (CIS)

- First episode of neurologic symptoms that **lasts at least 24 hours**
- Patient reported symptoms that suggest MS without confirmed diagnosis
- Symptoms may include unilateral optic neuritis, partial myelopathy, or focal syndromes

Relapsing-Remitting Multiple Sclerosis (RRMS)

- **Most common** (85 – 90% at onset)
- RRMS consists of clearly defined worsening of symptoms, called relapses, following by partial or full improvement of symptoms
- Relapses can last days to weeks
- Remission periods can last weeks to years
 - Goal of DMT is to keep patients in remission and to avoid relapses

THE 4 SUBTYPES OF MULTIPLE SCLEROSIS

Secondary Progressive Multiple Sclerosis (SPMS)

- Progressive course that follows RRMS
- Symptoms slowly worsen with variable remission and relapses
- 65% of patients convert 15-20 years after disease onset

Primary Progressive Multiple Sclerosis (PPMS)

- 10 – 15% at onset
- Neurologic disability that *steadily worsens from disease onset*
- No distinct periods of remission or relapse
- Diagnosis can only be made from disease history
- Average age of onset: 40 years old

GOALS OF THERAPY

- ✓ Reduce number, severity and duration of relapses
- ✓ Slow disease progression
- ✓ Improve quality of life



There is currently no cure for MS

NON-PHARMACOLOGICAL MEASURES

Non-pharmacological measures only **provide symptomatic improvement**

Exercising can improve balance, coordination, muscle tone and strength (e.g. pelvic floor exercises for bladder symptoms)

Avoiding excessive heat or cold exposure as symptoms are exacerbated by extremes of temperature

Physiotherapy, stretching, and massage therapy can provide relief for spasticity and gait improvement

PHARMACOLOGICAL THERAPY

- There is no cure for MS, but medications are available that may reduce relapses and delay progression of the disease
 - DMTs work by reducing inflammation associated with relapse periods
 - Due to the nature of PPMS and SPMS, DMTs have less utility than in RRMS unless the patient has frequent relapses, with some exceptions:
 - Ocrelizumab is approved in Canada for initial therapy in patients with PPMS with active disease
 - Siponimod is indicated for patients with SPMS to delay progression of physical disability
- DMTs are typically prescribed by specialists due to the risks associated with them

PHARMACOLOGICAL THERAPY

FIRST-LINE DISEASE-MODIFYING THERAPIES (lower-efficacy therapies)

Interferon beta (Injection)

- Standard first-line treatment that reduces conversion from CIS to definite MS
- Reduces relapse rates and the number and size of MRI-detectable lesions

Glatiramer acetate (Injection)

- Reduces relapse rates (may reduce conversion), and reduces MRI-detectable lesions

PHARMACOLOGICAL THERAPY

FIRST-LINE DISEASE-MODIFYING THERAPIES (lower-efficacy therapies)

Dimethyl fumarate (Oral)

- Activates nuclear factor-like 2 pathway that is involved in cellular responses to oxidative stress
- Reduces the number and rate of relapses and number of new/enlarging brain lesions

Teriflunomide (Oral)

- Immunomodulator that reduces B- and T-cell activity by inhibiting dihydroorotate dehydrogenase-mediated pyrimidine synthesis
- Reduces relapse rates and disease progression
- Favourable safety profile with no known cases of PML reported
- Can affect sperm and cause teratogenic effects, may last for 2 years after stopping

PHARMACOLOGICAL THERAPY

FIRST-LINE DISEASE-MODIFYING THERAPIES (moderate-efficacy therapies)

Siponimod (Oral)

- Sphingosine-1-phosphate receptor agonist used **first-line in SPMS** to delay disability progression
- Contraindicated in patients with CYP2C9*3*3 genotype

Ozanimod (Oral)

- Sphingosine-1-phosphate receptor agonist used in patients with RRMS
- Cannot be used in combination with MAOIs and patient should avoid tyramine-rich foods due to risk of hypertensive crisis

Ponesimod (Oral)

- Sphingosine-1-phosphate receptor agonist used in patients with RRMS

PHARMACOLOGICAL THERAPY

SECOND-LINE DISEASE-MODIFYING THERAPIES

These are considered **HIGHER-EFFICACY THERAPIES**, with the exception of fingolimod*

Alemtuzumab (Infusion)

- Monoclonal antibody that binds to CD52 on lymphocytes
- Reduces disability and relapse in RRMS compared to interferon-beta
- Typically reserved for very active disease

Cladribine (Oral)

- Slowly depletes then reconstitutes B and T lymphocytes
- Reduces relapse rate, disability progression and MRI disease activity
- Given for very short courses over 2 years, then patient stops

Fingolimod (Oral)

* Efficacy was found to be similar to that of lower-efficacy agents

- A sphingosine-1-phosphate receptor agonist
- Reduces frequency of relapses and inflammatory activity in RRMS
- Precautions needed for serious adverse drug events, so typically reserved for very active disease



Higher-efficacy agents should be first choice for all patients with more severe/aggressive disease presentation (e.g. frequent or disabling relapses, incomplete recovery after a relapse, high lesion burden on imaging, early deterioration of function)

PHARMACOLOGICAL THERAPY

SECOND-LINE DISEASE-MODIFYING THERAPIES

These are considered **HIGHER-EFFICACY THERAPIES**

Ocrelizumab (Infusion)

- Used **first-line for PPMS** or second-line in RRMS
- Humanized monoclonal antibody that depletes CD-20 B cells
- Decreases clinical and MRI progression

Natalizumab (Infusion)

- An alpha-4 integrin antagonist/adhesion molecule inhibitor
- Reduces relapse rates and gadolinium-enhancing lesions in RRMS
- Higher PML risk – reserved for very active disease



Higher-efficacy agents should be first choice for all patients with more severe/aggressive disease presentation (e.g. frequent or disabling relapses, incomplete recovery after a relapse, high lesion burden on imaging, early deterioration of function)

PHARMACOLOGICAL THERAPY

SECOND-LINE DISEASE-MODIFYING THERAPIES

These are considered **HIGHER-EFFICACY THERAPIES**

Ofatumumab (Injection)

- Monoclonal antibody that targets CD-20 B cells used in RRMS
- Shown to significantly reduce relapses, slow disability progression, and decrease new lesion formation
- No cases of PML reported during trials but still labeled as a risk



Higher-efficacy agents should be first choice for all patients with more severe/aggressive disease presentation (e.g. frequent or disabling relapses, incomplete recovery after a relapse, high lesion burden on imaging, early deterioration of function)

PHARMACOLOGICAL THERAPY

ADJUNCTIVE THERAPIES

Fampridine (4-aminopyridine)

- Selectively blocks K⁺ channels to increase strength of action potentials; improves walking ability (walking speed and ambulatory function)

PHARMACOLOGICAL THERAPY

RELAPSES

IV Option

- Methylprednisolone 1g IV for 3-5 days to help shorten duration of the relapse +/- oral prednisone taper

Oral Option

- Prednisone 1250mg daily for 3-5 days +/- oral prednisone taper was found to be non-inferior to IV methylprednisolone

Used in relapses only

- Steroids are used to control the inflammation causing the relapse, they have no effect on the course of the disease (they are not used as DMTs)

Severe relapses may require plasmapheresis (FYI)

PHARMACOLOGICAL THERAPY

FYI

SYMPTOMATIC MANAGEMENT

Neuropathic pain

Gabapentin, pregabalin

Fatigue

Modafinil, amphetamine,
lisdexamfetamine,
methylphenidate

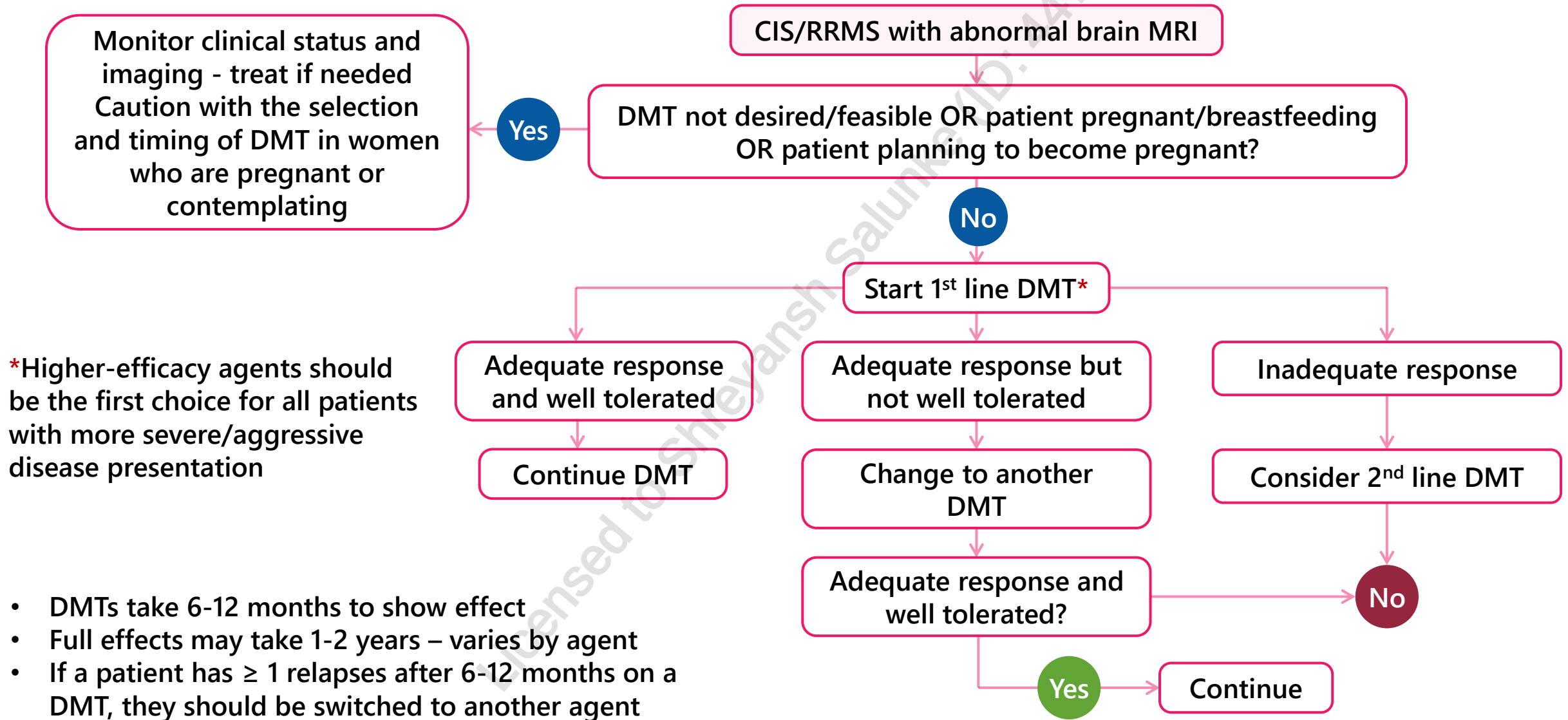
Spasms

Baclofen (oral or intrathecal),
cyclobenzaprine, tizanidine,
nabiximols, benzodiazepines

Bladder symptoms

Oxybutynin, mirabegron,
botulinum toxin (into bladder)

MANAGEMENT OF MULTIPLE SCLEROSIS



PREGNANCY AND BREASTFEEDING

Pregnancy

- Pregnancy is associated with decreased disease activity and lower relapse rates
- Immunosuppressive or immunomodulator treatment is NOT recommended
 - Teriflunomide, cladribine, and S1P receptor modulators are contraindicated
 - Patients using teriflunomide who are planning pregnancy should have a washout period with cholestyramine prior to conception
- Generally, immunomodulators should be discontinued 2-6 months prior to conception (in most cases, depending on the drug)
- Limited studies suggest glatiramer acetate and interferon beta are not associated with increased birth defects



Breastfeeding

- Avoid teriflunomide, cladribine, and mitoxantrone
- Breastfeeding may reduce the risk of relapse thus immunomodulator treatments are typically discontinued during this time
- No significant risks have been reported regarding maternal immunomodulator use during breastfeeding



DRUG TABLES

Class?	Drug?	Side effects?	Comments?
Immune modulators	Dimethyl fumarate	GI (abdominal pain, diarrhea, nausea, vomiting), flushing, increased liver enzymes, decreased lymphocytes, proteinuria, rare risk of PML Monitor CBC, liver function, urinalysis	Avoid live vaccines; administered PO
	Glatiramer	Anxiety, chest pain, flushing, hypertonia, joint pain, nausea, palpitations, weakness, post-injection reaction, injection site pain	No severe interactions stated; time to effect is 6 months; administered SC
	Interferon beta-1a	Elevated liver enzymes, decreased platelets, flu-like symptoms (chills, fever, myalgias), injection-site reactions; depression, fatigue, headache, pancytopenia (reduced blood cells), hypertension, tachycardia	Could reduce immune response to live vaccines; time to effect is 3 months; administered IM or SC
	Interferon beta 1-b		See above
	Peginterferon beta-1a		See above
Monoclonal Antibodies	Ocrelizumab	URTIs, nasopharyngitis, infusion reactions (HA, rash, fever, urticaria), PML (rare) Assess Hep B status prior to treatment. Give methylprednisolone 100 mg IV 30 min prior to infusion, also consider pre-medication with antihistamine and antipyretic	Avoid live vaccines during and post-treatment until B-cell levels normalize; administered IV; time to effect is 3 months



Consider using analgesics or antipyretics before injecting to minimize injection-site and flu-like reactions

DRUG TABLES

Class?	Drug?	Side effects?	Comments?
Pyrimidine synthesis inhibitors	Teriflunomide	Elevated hepatic enzymes (serious liver toxicity risk), diarrhea, alopecia, hypertension, neutropenia, paresthesia, rash. Monitor hepatic enzymes at baseline and monthly x 6 months. Wash-out protocol with cholestyramine, used to minimize exposure in patients who may have liver damage OR women who become pregnant while taking teriflunomide	Drug interactions: BRCP efflux transporter inhibitors (e.g. cyclosporine, eltrombopag, gefitinib) may increase teriflunomide exposure; decreased INR in patients on warfarin – monitor closely; CI concurrent use with leflunomide, pimecrolimus, tacrolimus Avoid live vaccines; time to effect is 3 months; CI in pregnancy

DRUG TABLES

Class?	Drug?	Side effects?	Comments?
Immune Modulators	Fingolimod	Brady-arrhythmias (monitor patient for 6h following 1 st dose), QT _c interval prolongation, infection, increased liver enzymes, macular edema, increased blood pressure with ongoing Tx, rare PML Baseline ECG and CBC, baseline liver enzymes and monitor Q monthly x 3 months, then 2x/year, reactivation of hepatitis B (ensure patient immune or vaccinate prior to starting), infections	Drug interactions: • CYP3A4, CYP2C9, and CYP4F2 inhibitors • QT_c-prolonging drugs • Avoid live vaccines • MAOi medications and tyramine-rich foods (O) Contraindications: • Patients with a CYP2C9*3*3 genotype (S) • Patients that have had an MI, unstable angina pectoris, or stroke/TIA in the last 6 months, patients with class III or IV heart failure • Pregnancy or patients wishing to become pregnant • Liver impairment (Child-Pugh Class B to C)
	Siponimod		
	Ozanimod		
	Ponesimod		

DRUG TABLES

Class?	Drug?	Side effects?	Comments?
Monoclonal antibodies	Natalizumab	Arthralgias, depression, fatigue, headache, infection, increased liver enzymes Progressive multifocal leukoencephalopathy (rare but fatal)	Increased drug levels (30%) when combined with interferon beta-1a; time to effect is 1 month; administered IV
Monoclonal antibodies	Alemtuzumab	Autoimmunity (e.g. anti-glomerular basement membrane disease, Graves' hyperthyroidism, thrombocytopenic purpura), serious infections, infusion reactions (pre-medicate with corticosteroid for first 3 days of tx), PML (rare but fatal), autoimmune hepatitis, hemophagocytic lymphohistiocytosis, cardiovascular reactions Monitor LFTs, CBC and CV function. Antiviral prophylaxis recommended (minimum of 1 month).	Avoid live vaccines; administered IV. Full benefit seen in 2 years.
Purine analogue	Cladribine	Lymphopenia and herpes zoster infections. Lymphocyte count must be normal before initiating and ≥ 800 cells/mm ³ before initiating in year 2.	Separate all other orals meds by minimum of 3 hours from cladribine; avoid live or live-attenuated vaccines during treatment and after discontinuation until white blood cell counts are within normal limits; screen for HIV, TB, hepatitis B and C prior to each course; check the Varicella Zoster Virus antibody status prior to treatment initiation; CI pregnancy, breastfeeding, CrCl <60 mL/min. Full benefit seen in year 2.

DRUG TABLES

Class?	Drug?	Side effects?	Comments?
Monoclonal Antibodies	Ofatumumab	Upper respiratory infections, injection-site reactions (both local and systemic), headache, fatigue, neutropenia, decreased IgM levels, hepatitis B reactivation (ensure patient titers are appropriate and/or revaccination prior to starting treatment)	Patients may wish to pre-medicate with an anti-pyretic and pain reliever prior to first dose Contraindicated in patients with active infections, malignancies, Hepatitis B, and severely immunocompromised patients No significant drug interactions aside from live vaccinations, which should be avoided prior to treatment. Non-live vaccinations should be given in the middle of the dosing interval
Potassium channel blockers	Fampridine	Increased risk of seizures (dose-related); avoid in renal impairment or history of seizures; do NOT exceed usual dose. Balance disorders, headache, nausea, paresthesia, urinary tract infection.	Drug interactions: Organic cation transporter (OCT2) inhibitors and substrates may decrease fampridine clearance and increase seizure risk. Administered PO; swallow whole with a glass of water; do not take with food Assess walking ability after 4 wk. If no benefit, discontinue drug. Advise patients: ➤ Not to take doses less than 12H apart ➤ Missed doses should not be taken late ➤ Omit missed dose and resume with the next scheduled dose

MONITORING PARAMETERS

Efficacy Endpoints			
Parameter	Monitoring Tools	Frequency	Degree of Change
Frequency of relapses	Physical exam, EDSS, MRI	At every neurology appointment	Prevent
New lesions	MRI	Every 6-24 months	
Walking speed	Timed 25-foot walk	At every neurology appointment	Stable
MS Symptoms	Physical exam, EDSS		Reduce

MONITORING PARAMETERS

Safety Endpoints					
Parameter	Monitoring tools	Frequency	Degree of Change	Timeframe	
PML	JCV levels	Baseline Every 6 months*	None	Duration of therapy	
Liver Function	LFTs	Baseline then annually, or if clinically indicated	Within normal range		
Neutropenia	CBC		Prevent		
Infection	Antibiotic use Patient exam	Every visit	Prevent or minimize		
Other adverse effects	Patient questionnaire				

TAKEAWAY

- **The most common form of MS is RRMS.** RRMS is likely to respond to immunomodulator therapy
- There is no cure for MS. Early treatment with an immunomodulator improves quality of life and long-term neurological outcomes
- **Higher-efficacy agents should be the first choice for all patients with more severe/aggressive disease presentation** (e.g. frequent or disabling relapses, incomplete recovery after a relapse, high lesion burden on imaging, early deterioration of function)
 - **Higher efficacy drugs:** natalizumab, ocrelizumab, ofatumumab, alemtuzumab, cladribine
- Vitamin D supplementation for all MS patients (min 1000 IU/day)
 - Although there is little evidence for benefit, they show other benefits and harm is generally minimal if taken in moderation.
 - Before initiating therapy in patients, it is important to compare therapies and understand their strengths and limitations
 - Most disease-modifying therapies are **not useful in patients with progressive disease without relapse**

CASE #1

KL is a 30-year-old female that is newly diagnosed with RRMS (mild) by her neurologist. Her symptoms include numbness and tingling of the extremities, an electric shock sensation when flexing her neck, pain, and fatigue. Her past medical history is significant for obesity, depression and anxiety. Her current medications include duloxetine 60mg daily, buspirone 10mg three times daily, and vitamin D3 1000 IU daily, as she is deficient. She states she is the only member of her family that has been diagnosed with MS. She does not smoke but drinks socially, approximately 1-2 drinks per week. Her neurologist wants your assistance to determine what DMT to start in KL

1. How many risk factors for MS does KL have?
 - a) 2
 - b) 3
 - c) 4
 - d) 5



CASE #1

KL is a 30-year-old female that is newly diagnosed with RRMS (mild) by her neurologist. Her symptoms include numbness and tingling of the extremities, an electric shock sensation when flexing her neck, pain, and fatigue. Her past medical history is significant for obesity, depression and anxiety. Her current medications include duloxetine 60mg daily, buspirone 10mg three times daily, and vitamin D3 1000 IU daily, as she is deficient. She states she is the only member of her family that has been diagnosed with MS. She does not smoke but drinks socially, approximately 1-2 drinks per week. Her neurologist wants your assistance to determine what DMT to start in KL.

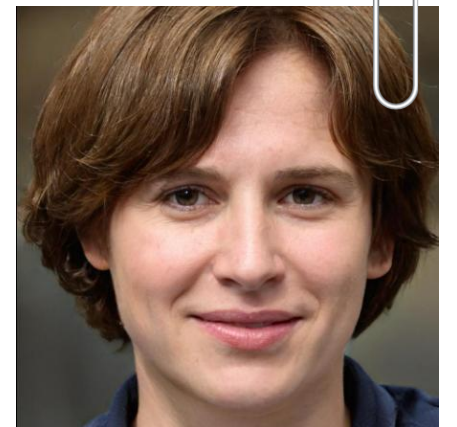
2. Which medication is the MOST appropriate DMT to treat KL's RRMS?
- a) Prednisone 1250mg daily x 5 days followed by a taper
 - b) Methylprednisolone 1g IV x 5 days, followed by a taper
 - c) Dimethyl fumarate 120mg BID x 7 days, followed by 240mg BID thereafter
 - d) Ofatumumab 20mg at weeks 0, 1, 2, and 4, followed by 20mg once monthly thereafter



CASE #1

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3. Which of the following vaccinations should be avoided in KL once she starts a DMT?
- a) Varicella
 - b) Shingles
 - c) COVID-19
 - d) Influenza



CASE #1

KL is a 30-year-old female that is newly diagnosed with RRMS (mild) by her neurologist. Her symptoms include numbness and tingling of the extremities, an electric shock sensation when flexing her neck, pain, and fatigue. Her past medical history is significant for obesity, depression and anxiety. Her current medications include duloxetine 60mg daily, buspirone 10mg three times daily, and vitamin D3 1000 IU daily, as she is deficient. She states she is the only member of her family that has been diagnosed with MS. She does not smoke but drinks socially, approximately 1-2 drinks per week. Her neurologist wants your assistance to determine what DMT to start in KL.

4. If KL had been diagnosed with PPMS, which of the following DMTs would be the MOST appropriate?

- a) Prednisone 1250mg daily x5 days followed by a taper
- b) Dimethyl fumarate 120mg BID x 7 days, followed by 240mg BID thereafter
- c) Fampridine 10mg every 12 hours
- d) Ocrelizumab 300mg every 14 days x 2 doses, followed by 600mg every 6 months thereafter



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CHANGE LOG

July 2025

- Formatting edits made throughout; no content changes made
- Legend updated

September 2025

- Slide 3: Updated legend
- Slide 29-30: Elaborated on monitoring parameters

November 2025

- Formatting and minor phrasing changes made

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 3: Updated legend
- Reformatted and expanded some slides
- Slides 14/15/16: Added 'lower-efficacy therapies' in title and removed note
- Slides 17/18: Added 'higher-efficacy therapies' to title and important note on their place in therapy
- Slide 18: Removed mention of ublituximab since not on Canadian market yet
- Slide 23: Updates to algorithm
- Slide 32: Updated the takeaway slide
- Slides 33-36: Added 'mild' to case, no changes in questions